STPM Physics · Semester 1 · Chapter 1

Physical Quantities

1.1 Physical Quantities & Units

Definitions & laws

- **Base quantities** are physical quantities that are independent and cannot be defined in terms of other quantities. The seven SI base quantities and units are: length (metre, m), mass (kilogram, kg), time (second, s), electric current (ampere, A), thermodynamic temperature (kelvin, K), amount of substance (mole, mol) and luminous intensity (candela, cd).
- **Derived quantities** are quantities defined in terms of base quantities by multiplication or division; their units are derived units (e.g. N = kg m s^{-2}, J = kg m^2 s^{-2}, W = kg m^2 s^{-3}, Pa = kg m^{-1} s^{-2}).
- **Dimension** of a physical quantity is the way it is expressed in terms of the base quantities, written in square brackets, e.g. [L], [M], [T].
- **Homogeneity principle**: every term in a physically correct equation must have the same dimensions (and units).
- Prefixes: tera T 10^{12}, giga G 10^{9}, mega M 10^{6}, kilo k 10^{3}, deci d 10^{-1}, centi c 10^{-2}, milli m 10^{-3}, micro μ 10^{-6}, nano n 10^{-9}, pico p 10^{-12}.
- Dimensionless quantities (pure numbers, angles in radian, ratios, refractive index, strain) have no unit.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Speed or velocity (dimensions)	$[v] = [L][T]^{-1}$	[L] = length; [T] = time; SI unit m s^{-1}
Acceleration (dimensions)	$[a] = [L][T]^{-2}$	[L] = length; [T] = time; SI unit m s^{-2}
Force (dimensions)	$[F] = [M][L][T]^{-2}$	[M] = mass; [L] = length; [T] = time; SI unit N
Energy and work (dimensions)	$[M][L]^2[T]^{-2}$	SI unit J
Power (dimensions)	$[M][L]^2[T]^{-3}$	SI unit W
Pressure (dimensions)	$[M][L]^{-1}[T]^{-2}$	SI unit Pa (N m^{-2})
Density (dimensions)	$[M][L]^{-3}$	SI unit kg m^{-3}

Quantity / law	Formula	Symbols, units & conditions
Frequency (dimensions)	$[T]^{-1}$	SI unit Hz
Momentum (dimensions)	$[M][L][T]^{-1}$	SI unit kg m s^{-1}

Unit conversions

Quantity	Conversion	Working
Speed	1 km h^{-1} = $\frac{1}{3.6}$ m s^{-1} ≈ 0.278 m s^{-1}	1 km h^{-1} = $\frac{1000\text{ m}}{3600\text{ s}}$; 3600 ÷ 1000 = 3.6. So km h^{-1} ÷ 3.6 gives m s^{-1}, and m s^{-1} × 3.6 gives km h^{-1} (e.g. 72 km h^{-1} = 20 m s^{-1})
Density	1 g cm^{-3} = 10^3 kg m^{-3}	1 g = 10^{-3} kg and 1 cm^3 = 10^{-6} m^3, so $\frac{10^{-3}}{10^{-6}}$ = 10^3
Area	1 cm^2 = 10^{-4} m^2	1 cm = 10^{-2} m, so 1 cm^2 = $(10^{-2}\text{ m})^2$ = 10^{-4} m^2
Volume	1 cm^3 = 10^{-6} m^3	1 cm = 10^{-2} m, so 1 cm^3 = $(10^{-2}\text{ m})^3$ = 10^{-6} m^3

How to answer (MPM style)

- To check an equation: write the dimensions (or SI base units) of every term, then compare both sides; state "the equation is homogeneous / dimensionally consistent" or "not homogeneous, so it is incorrect".
- To find the unit of a constant: make the constant the subject, substitute SI units of each quantity, simplify to base units, then name the unit if it has a special name.
- Give final units in SI with correct prefix conversion before substituting into any formula.
- A conclusion must be worded: homogeneity is necessary but not sufficient for correctness.

Section A traps

Trap: Dimensional analysis can prove an equation wrong but cannot prove it correct: dimensionless factors such as ½, 2 or π are not tested, and a wrong equation can still be homogeneous.

Trap: Unconverted prefixes: 5 cm^2 is 5×10^{-4} m^2, not 5×10^{-2} m^2; 3 g cm^{-3} is 3×10^3 kg m^{-3}.

Trap: Kilogram, not gram, is the SI base unit of mass; newton, joule and watt are derived units, not base units.

Tricks

Trick: Replace every quantity by its base-unit string and cancel; if the powers differ on the two sides, eliminate that option at once.

> **Trick:** The argument of an exponential, sine or logarithm function must be dimensionless, so the argument has no unit, which fixes the unit of any constant inside it.

1.2 Scalars and Vectors

Definitions & laws

- **Scalar quantity** is a quantity that has magnitude only (e.g. distance, speed, mass, time, energy, temperature).
- **Vector quantity** is a quantity that has both magnitude and direction (e.g. displacement, velocity, acceleration, force, momentum).
- **Resultant** is the single vector that has the same effect as the vectors it replaces.
- Vectors are added by the triangle law (head to tail) or the parallelogram law (tail to tail, resultant is the diagonal from the common tail).
- **Vector subtraction**: $A - B = A + (-B)$, where $-B$ has the same magnitude as B but opposite direction.
- **Resolution** of a vector is replacing it by two perpendicular components that together have the same effect.
- **Unit vector** has magnitude 1 and gives direction only; a vector is written $A = A_x i + A_y j$.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Vector components	$A_x = A\cos\theta$; $A_y = A\sin\theta$	A = magnitude of the vector; θ = angle between A and the x-axis; A_x, A_y = components along x and y
Magnitude and direction	$A = \sqrt{A_x^2 + A_y^2}$; $\tan\theta = \frac{A_y}{A_x}$	A = magnitude; A_x, A_y = components; θ = direction angle from the x-axis
Resultant of two vectors	$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$	R = magnitude of resultant; A, B = magnitudes of the vectors; θ = angle between them (tail to tail)
Resultant of perpendicular vectors	$R = \sqrt{A^2 + B^2}$	A, B = magnitudes of the two perpendicular vectors; R = magnitude of resultant
Resultant of equal vectors	$R = 2A\cos\frac{\theta}{2}$	A = magnitude of each vector; θ = angle between them; R = magnitude of resultant
Sine rule	$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma}$	a, b, c = sides of the vector triangle; α, β, γ = angles opposite a, b, c
Cosine rule	$c^2 = a^2 + b^2 - 2ab\cos\gamma$	a, b, c = sides of the vector triangle; γ = angle opposite c
Resultant range	$\lvert A - B\rvert \le R \le A + B$	A, B = magnitudes of the two vectors; R = magnitude of resultant

How to answer (MPM style)

- Draw a labelled vector diagram (triangle or parallelogram) with arrows and angles before calculating.
- Resolve each vector along two chosen perpendicular axes, sum the components (ΣF_x, ΣF_y), then state $R = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$ and the angle $\theta = \tan^{-1}\frac{\Sigma F_y}{\Sigma F_x}$ with respect to a stated axis.
- State the direction of the resultant, as an angle and its reference axis, as well as its magnitude with unit.
- For a change in a vector (e.g. $\Delta v = v - u$), draw u reversed and add.

Section A traps

Trap: Using θ from the axis in $R^2 = A^2 + B^2 + 2AB\cos\theta$: θ must be the angle between the two vectors tail to tail; head-to-tail triangle uses the cosine rule with a minus sign and the exterior angle.

Trap: Sine and cosine swapped: the component adjacent to the stated angle uses cos θ; if the angle is measured from the vertical or from an incline surface, the cos and sin roles interchange.

Trap: Distance and speed are scalars; displacement and velocity are vectors. A body in uniform circular motion has constant speed but changing velocity.

Trap: Magnitude of the vector difference of two equal-magnitude perpendicular vectors is $\sqrt{2}$ times the magnitude, not zero and not 2 times.

Tricks

Trick: Test limiting angles: θ = 0° gives R = A + B, 90° gives $\sqrt{A^2 + B^2}$, 180° gives $|A - B|$; eliminate options outside this range.

Trick: Three-four-five and 5-12-13 triples save time in perpendicular resultants.

1.3 Uncertainties in Measurements

Definitions & laws

- **Accuracy** is the closeness of a measured value to the true (accepted) value; it is limited by systematic error.
- **Precision** is the closeness of repeated measurements to one another (the spread of readings); it is limited by random error and indicated by a small uncertainty.
- **Systematic error** is an error that makes all readings differ from the true value by the same amount in the same direction (e.g. zero error, wrongly calibrated instrument, parallax by a consistently wrong sight); it cannot be reduced by averaging, but can be corrected by checking zero or calibrating.
- **Random error** is an error that causes readings to scatter above and below the true value in an unpredictable manner (e.g. reaction time, parallax by varying sight); it is reduced by taking many readings and calculating the mean.

- **Absolute uncertainty** Δx is the uncertainty in a measurement, with the same unit; **fractional uncertainty** is $\frac{\Delta x}{x}$; **percentage uncertainty** is $\frac{\Delta x}{x} \times 100\%$.
- Uncertainty of a single reading: analogue instrument ± half of the smallest division (some scales: ± one smallest division); digital instrument ± one smallest digit.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Mean of repeated readings	$\bar{x} = \frac{\Sigma x}{n}$	$\bar{x}$ = mean value; x = individual reading; n = number of readings
Uncertainty from repeated readings	$\Delta x = \frac{x_{max} - x_{min}}{2}$	Δx = uncertainty; x_{max}, x_{min} = largest and smallest readings
Sum or difference	$Z = A \pm B$; $\Delta Z = \Delta A + \Delta B$	ΔZ, ΔA, ΔB = absolute uncertainties, which add
Product or quotient	$Z = AB$ or $\frac{A}{B}$; $\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$	$\frac{\Delta Z}{Z}$, $\frac{\Delta A}{A}$, $\frac{\Delta B}{B}$ = fractional uncertainties, which add
Power	$Z = A^n$; $\frac{\Delta Z}{Z} = \lvert n \rvert \frac{\Delta A}{A}$	n = power; $\frac{\Delta Z}{Z}$, $\frac{\Delta A}{A}$ = fractional uncertainties
General combination	$Z = A^m \frac{B^n}{C^p}$; $\frac{\Delta Z}{Z} = \frac{\lvert m \rvert \Delta A}{A} + \frac{\lvert n \rvert \Delta B}{B} + \frac{\lvert p \rvert \Delta C}{C}$	m, n, p = powers; $\frac{\Delta A}{A}$, $\frac{\Delta B}{B}$, $\frac{\Delta C}{C}$ = fractional uncertainties
Constant factor	$Z = kA$; $\Delta Z = k\Delta A$	k = exact number; it does not add to the fractional uncertainty

- Gradient of a graph: uncertainty = (gradient of best-fit line − gradient of worst acceptable line); similarly for intercepts.

How to answer (MPM style)

- State the result as (x ± Δx) unit, with Δx to 1 significant figure (2 if it begins with 1) and x to the same decimal place as Δx.
- For a derived quantity: write the relation, state the combining rule used, substitute fractional uncertainties, then convert to the absolute uncertainty.
- For "suggest how to reduce error": random error, take repeated readings and find the mean; systematic error, correct the zero error or recalibrate; reading error, use an instrument with smaller scale division or measure a larger quantity (e.g. time 20 oscillations).
- On graphs use a straight line of best fit with points balanced, large triangle for gradient.

Section A traps

Trap: Subtracting uncertainties: uncertainties always add, even when quantities are subtracted or divided.

Trap: Forgetting the power: for $Z = A^2B$, the contribution of A is $\frac{2\Delta A}{A}$; for a radius used as r^2, percentage uncertainty doubles.

Trap: Averaging repeated readings does not remove a zero error; a precise instrument can be inaccurate, and a set of readings with small spread can still have a large systematic error.

Trap: Adding absolute uncertainties for a product or adding fractional uncertainties for a sum: match the rule to the operation.

Tricks

Trick: Convert everything to percentage uncertainty first when the formula contains products, quotients or powers; the answer is a simple weighted sum.

Trick: Quick check: the combined uncertainty is always larger than the largest single contribution and never larger than the sum of all contributions.